

MACKAY AP, Australia

WMO: 953670

Lat: 21.1706S

Lon: 149.1794E

Elev: 16

StdP: 14.69

Time Zone: 10.00 (E10)

Period: 96-15

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	44.7	47.5	36.1	31.2	55.7	39.5	35.6	55.1	22.4	70.4	21.0	70.1	6.2	270	0.471

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	10.7	89.4	77.3	87.8	76.4	86.5	75.9	80.1	85.1	79.0	84.3	78.1	83.7	14.6	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
78.9	150.1	82.8	77.5	143.0	82.4	76.4	137.7	81.9	43.9	85.1	42.6	84.4	41.7	83.8	89.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.8	23.2	21.1	40.1	93.6	3.0	2.0	37.9	95.0	36.2	96.2	34.5	97.3	32.3	98.8
			38.0	81.8	2.7	1.9	36.0	83.2	34.4	84.3	32.9	85.4	30.9	86.8
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	72.4	80.2	79.7	77.7	73.8	68.5	64.4	62.3	63.9	68.6	73.7	77.0	79.4	(6)
	DBStd	7.20	2.36	2.14	2.56	2.82	3.90	4.66	4.47	3.99	3.77	3.42	2.85	2.79	(7)
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
	HDD65	267	0	0	0	0	16	66	106	68	11	0	0	0	(9)
	CDD50	8171	935	830	858	715	575	433	381	429	559	736	810	910	(10)
	CDD65	2964	470	410	393	266	126	49	22	33	119	271	360	445	(11)
	CDH74	23659	4795	4007	3273	1496	378	70	29	116	614	1687	2862	4332	(12)
	CDH80	5477	1371	1055	649	146	10	3	0	4	65	254	635	1285	(13)
(14) Wind	WSAvg	11.3	12.5	12.8	13.7	11.8	10.8	9.6	9.2	9.4	10.2	11.9	11.7	11.8	(14)
(15) Precipitation	PrecAvg	65.5	11.7	13.4	9.9	6.1	3.7	2.2	1.3	1.3	0.7	1.9	4.3	7.7	(15)
	PrecMax	111.0	45.0	41.7	32.8	22.5	15.3	14.4	8.2	11.7	3.0	7.9	19.0	51.6	(16)
	PrecMin	33.6	1.0	1.5	0.6	0.8	0.4	0.1	0.0	0.0	0.0	0.0	0.0	0.6	(17)
	PrecStd	20.7	10.1	9.8	7.6	4.7	2.8	2.4	1.4	2.0	0.7	1.8	3.9	8.3	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	90.9	90.8	88.0	85.6	81.4	78.5	77.0	80.0	84.6	87.2	89.2	91.7	(19)
		MCWB	79.0	79.3	77.1	74.9	70.9	69.5	65.9	66.0	70.2	73.4	74.8	76.6	(20)
	2%	DB	88.7	87.9	86.2	83.0	79.0	75.8	74.3	77.0	81.8	84.8	86.7	89.3	(21)
		MCWB	77.9	77.6	76.1	73.7	69.8	68.4	65.9	66.4	69.6	72.5	74.0	76.4	(22)
	5%	DB	87.1	86.4	84.9	81.6	77.4	74.2	72.7	74.7	79.4	82.8	85.1	87.5	(23)
		MCWB	77.0	77.2	75.3	72.9	69.2	67.4	64.8	65.5	68.3	71.8	73.3	75.6	(24)
	10%	DB	85.7	85.2	83.5	80.4	76.2	72.6	71.2	72.9	77.4	80.9	83.6	85.9	(25)
		MCWB	76.4	76.8	74.6	72.2	68.8	66.2	63.8	64.9	67.4	70.3	72.3	74.7	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	82.0	81.8	80.2	78.2	75.4	72.7	70.9	71.1	73.7	77.3	77.3	80.7	(27)
		MCDB	86.4	87.5	83.9	82.0	77.9	76.3	73.9	75.0	80.4	81.5	85.7	86.4	(28)
	2%	WB	80.1	80.1	78.4	76.4	73.1	70.8	68.7	69.3	71.9	75.1	76.0	78.5	(29)
		MCDB	85.3	84.7	83.3	80.3	76.1	74.1	72.2	73.9	78.0	80.8	84.0	85.3	(30)
	5%	WB	78.9	78.9	77.3	74.9	71.4	69.4	67.2	67.9	70.7	73.7	74.9	77.4	(31)
		MCDB	84.5	83.9	82.3	79.4	75.5	72.6	70.8	72.8	76.8	80.3	82.5	84.3	(32)
	10%	WB	77.9	78.0	76.2	73.7	70.1	67.9	65.8	66.4	69.3	72.3	73.8	76.4	(33)
		MCDB	83.8	83.3	81.3	78.5	74.6	71.4	69.5	71.4	75.5	79.4	81.3	83.3	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	10.7	10.3	11.2	13.0	15.3	15.8	18.5	19.2	18.0	14.8	12.7	12.3	(35)
		MCDBR	12.9	12.8	12.7	14.5	16.0	15.7	18.4	20.7	20.9	16.9	14.6	14.7	(36)
	5% WB	MCWBR	5.1	5.5	5.7	7.5	8.8	9.4	10.9	11.9	10.4	7.4	5.7	5.3	(37)
		MCWBR	10.5	10.9	10.6	12.4	12.5	12.0	13.4	16.6	16.5	14.5	12.9	12.9	(38)
(40) Clear-Sky Solar Irradiance	taub		0.405	0.397	0.385	0.362	0.344	0.332	0.317	0.325	0.355	0.384	0.396	0.400	(40)
		taud	2.509	2.534	2.552	2.599	2.606	2.630	2.656	2.618	2.522	2.466	2.469	2.492	(41)
	Ebn at Noon		298	297	292	287	280	278	287	295	297	297	299	300	(42)
		Edh at Noon	36	35	33	29	27	25	25	28	33	37	38	37	(43)
(44) All-Sky Solar Radiation	RadAvg		2006	1887	1742	1544	1330	1179	1335	1605	1945	2147	2188	2178	(44)
	RadStd		150	207	166	150	85	116	102	112	136	125	254	230	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (8 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page